

An Internship Report
Of
Product Engineering

Submitted

To

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Engineering**

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**In partial fulfilment of the requirement of
degree of**

B-Tech (CSE)



By

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CANDIDATE'S DECLARATION

I, Anubhav Sachan, do hereby declare that the internship report titled "**Product Engineering**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

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I express my gratitude to my parents for their blessings.

Signature: 

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3. INTERNSHIP COMPLETION CERTIFICATE



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Product Engineering Internship undertaken at **MarTechAdda** as part of the Bachelor of Technology programme in Computer Science and Engineering. The internship focused on practical software product development, with major contributions to a **Customer Relationship Management (CRM) platform** and a standalone billing and invoicing product named **Ledgerly**.

The internship provided hands-on exposure to full-stack product engineering, including frontend development, backend integration, database interaction, financial workflow implementation, authentication, security, and collaborative software development. The work progressed from an initial learning phase into practical development, independent module ownership, cross-team integration, and product implementation.

The primary project involved the development of the CRM's **Escrow, Milestone, and Finance module**, designed to provide structured visibility into transactions, revenue, refunds, and payouts. In parallel, work was carried out on Ledgerly, a billing platform supporting invoices, estimates, client management, tax breakdowns, and printable financial documents.

4.2 Organization Profile

MarTechAdda is a technology and product-focused organization involved in the development of software solutions for client engagement, financial workflows, CRM operations, billing, and related business processes.

During the internship, the organization provided a practical product-development environment in which development activities were performed collaboratively using modern software engineering practices. The internship involved working with product-team members on CRM development, financial modules, billing workflows, backend integration, security issues, and application-level improvements.

The internship was supervised by **Ritesh Kumar**, who provided guidance throughout the internship period.

4.3 Problem Statement

As the organization's client and product requirements grew, financial operations such as transaction tracking, revenue management, refunds, and payouts required a more structured and auditable approach.

Previously, financial information within the CRM was not represented through a unified workflow, resulting in visibility gaps for transaction status, refunds, and payouts. Refund status changes also required a clearly defined lifecycle so that each stage could be tracked systematically.

Similarly, billing and invoicing operations required a dedicated product capable of managing invoices, estimates, clients, tax calculations, and printable documents. Manual handling of these activities could make quotation-to-invoice conversion, tax breakdowns, and client ledger management inefficient and error-prone.

The internship therefore focused on developing and improving software modules that could provide structured financial workflows, better data visibility, improved application security, and a more organized billing process.

4.4 Project Objectives

The major objectives of the internship were:

- To gain practical experience in product-oriented software engineering.
- To strengthen knowledge of React.js, TypeScript, Node.js, and Express.js.
- To understand and implement frontend interfaces for real-world business applications.
- To develop and contribute to the CRM Finance module.
- To implement transaction, revenue, refund, and payout management interfaces.
- To design a structured and auditable refund lifecycle.
- To understand backend and database integration in a collaborative development environment.
- To identify and resolve application-level security vulnerabilities.
- To develop billing and invoicing functionality for the Ledgerly platform.
- To implement invoice and quotation status workflows.
- To implement quotation-to-invoice conversion.
- To develop client management and financial information interfaces.
- To implement CGST/SGST tax breakdowns and printable financial documents.
- To apply Git and GitHub workflows for collaborative development.
- To gain experience in debugging, integration, state management, and data consistency issues.

4.5 Scope of the Project

The scope of the internship covered multiple areas of product engineering.

CRM Finance Module

The CRM Finance module included:

- Transaction Dashboard
- Revenue Dashboard
- Refund Tracking
- Payout Tracking
- Cash-flow and expenditure tracking
- Customer credit tracking
- Refund lifecycle management

The Refund Tracking module followed a structured lifecycle:

Requested → Under Review → Approved → Processing → Completed

with a separate **Rejected** branch.

CRM Integration and Security

The internship also covered:

- Express.js backend integration
- Database model integration
- Global state management
- Synchronization issue resolution
- Route protection
- Authorization-related security fixes
- Resolution of a privilege-escalation vulnerability
- Development database isolation and data-integrity improvements

Ledgerly Billing Software

The scope of Ledgerly included:

- Invoice management
- Estimate/quotation management
- Client directory
- Catalog profiles

- Quotation-to-invoice conversion
- Invoice status management
- CGST/SGST tax breakdown
- Printable financial documents
- Client ledgers
- Transaction records
- Activity timelines
- Contact information
- Server-side pagination

A personal developer portfolio was also developed to showcase the work completed during the internship.

4.6 Technologies and Tools Used

The following technologies and tools were used during the internship:

Frontend Development

- React.js
- TypeScript
- Tailwind CSS
- Vanilla CSS
- HTML/CSS principles
- Responsive UI development

Backend Development

- Node.js
- Express.js

Database and ORM

- Prisma ORM
- PostgreSQL
- MongoDB integration during CRM development

Authentication and Security

- Passport.js
- Google OAuth 2.0

- Protected routes
- Server-side authorization
- Role-based access considerations

Development and Collaboration

- Git
- GitHub
- Branch management
- Pull requests
- Code review
- Debugging and integration workflows

Development Assistance

- LLM-based development tools for code generation, debugging, refactoring, and development assistance

4.7 System Architecture

The projects developed during the internship followed a layered product-development approach.

Presentation Layer

The frontend was developed using React.js and TypeScript. Tailwind CSS and CSS were used to create responsive and reusable user interfaces.

Application Layer

The application layer handled business workflows such as:

- Transactions
- Revenue
- Refunds
- Payouts
- Invoices
- Estimates
- Client management

Backend Layer

Node.js and Express.js were used to provide server-side application functionality and integrate frontend requests with backend services and database models.

Authentication and Authorization Layer

Authentication was supported through Passport.js and Google OAuth 2.0. Protected routes and authorization mechanisms were also addressed during development.

Database Layer

Database models were integrated with the backend to store and retrieve financial, client, transaction, and billing-related information.

Version Control Layer

Git and GitHub were used for:

- Branch management
- Code collaboration
- Pull requests
- Integration
- Code review
- Version tracking

The overall development flow can be represented as:

**User Interface → React Application → Express.js Backend → Business Logic
→ Database**

with authentication and authorization applied to protected application functionality.

4.8 Methodology

The internship followed a six-week progression from technical learning to independent development and product integration.

Week	Module	Description
-------------	---------------	--------------------

- | | | |
|---|----------------------------|---|
| 1 | Development Foundations | Git/GitHub workflows, repository structure, branch management, pull requests, and introduction to the development workflow. |
| 2 | MERN/Full-Stack Foundation | Strengthening knowledge of React, Node.js, Express.js, database integration, and LLM-assisted development workflows. |
| 3 | Sample UI Designing | Development of responsive Blog and Resources page designs with category filters, archive layouts, search functionality, and improved readability. |
| 4 | CRM Finance Module | Development of the Transaction, Revenue, Refund, and Payout screens, with primary ownership of the Transaction Dashboard and refund workflow. |
| 5 | CRM Integration & Security | Cross-team integration, backend/database synchronization, global state management, route protection, security vulnerability resolution, and development database isolation. |
| 6 | Ledgerly & Portfolio | Development of the Ledgerly billing workspace, invoices, estimates, client management, tax documents, quotation-to-invoice conversion, pagination, and personal portfolio. |

Week 1–2: Learning Phase

The initial two weeks focused on strengthening software development fundamentals and establishing consistent development practices.

Git and GitHub were used to understand branch management, repository organization, pull requests, and collaborative development. The MERN/full-stack development foundation was strengthened through practical work with React.js, Node.js, Express.js, and database integration.

LLM-based tools were also used as development assistance for activities such as code generation, debugging, refactoring, and understanding implementation approaches.

Week 3: Sample Designing

The third week focused on converting design requirements into responsive frontend interfaces.

A **Blog Page** was developed with a premium responsive layout, readability-focused typography, image areas, and category filtering.

A **Resources Page** was also developed with an archive-style structure for guides, whitepapers, and downloadable resources, including search and category-selection functionality.

Week 4: CRM Finance Module

During the fourth week, major work was performed on the CRM Finance module.

Four major screens were developed:

1. Transaction Dashboard
2. Revenue Dashboard
3. Refund Tracking
4. Payout Tracking

Primary ownership was taken for the **Transaction Dashboard**, particularly the ledger view responsible for representing cash flow, expenditures, and customer credit.

The Refund Tracking system was designed around the following lifecycle:

Requested → Under Review → Approved → Processing → Completed

with a **Rejected** branch.

This provided a structured and traceable workflow for managing refund requests.

Week 5: Cross-Team Integration and Security

The fifth week focused on integrating the wider CRM application and collaborating with other members of the product team.

Work included connecting the Express.js server with database models, resolving synchronization issues, and structuring global state across the application.

During this phase, two important security issues were identified and resolved:

- Unguarded routes that could expose unauthorized data.
- A privilege-escalation vulnerability caused by an editable frontend role field.

The development environment also experienced data-integrity problems caused by the use of a shared development database. This issue was addressed by moving development work to an isolated local environment.

Week 6: Ledgerly Billing Software

The final week focused on the development of **Ledgerly**, a billing and invoicing platform.

The workspace included:

- Invoices
- Estimates
- Client directories
- Catalog profiles

Invoice status transitions were implemented using:

Draft → Sent → Approved → Paid

Quotation-to-invoice conversion was also implemented to simplify the billing workflow.

The billing system included CGST/SGST tax breakdowns and printable financial document interfaces. An eight-tab client profile provided access to client-related information such as ledgers, transaction records, activity timelines, and contacts.

Server-side pagination was implemented with **15 items per page** to improve interface performance when handling larger datasets.

A personal developer portfolio was also developed during this phase.

4.9 Expected Outcomes

The internship was expected to provide and resulted in practical experience in:

- Full-stack product engineering.
- React.js and TypeScript development.
- Responsive user-interface development.
- Backend development using Node.js and Express.js.
- Database integration and data management.
- CRM financial workflow development.

- Transaction and revenue management.
- Refund lifecycle implementation.
- Payout tracking.
- Billing and invoicing software development.
- Quotation-to-invoice automation.
- GST-oriented tax breakdown and printable document generation.
- Client profile and ledger management.
- Server-side pagination.
- Authentication and authorization.
- Identification and resolution of security vulnerabilities.
- Git/GitHub-based collaborative development.
- Debugging and integration of real-world software issues.

By the end of the internship, the work provided practical exposure to taking software features from initial design and development through integration, debugging, security improvement, and functional product delivery.

4.10 Certificate of Completion

The Internship Completion Certificate issued by MarTechAdda is attached on the following page.

The internship covered practical work in CRM development, financial workflow implementation, backend integration, security improvements, billing software development, and related product engineering activities.

5. BIBLIOGRAPHY / REFERENCES

You can replace the old generic frontend references with references relevant to the actual technologies used:

1. React Documentation, React.js documentation and development concepts.
2. Node.js Documentation, Node.js runtime and server-side development.
3. Express.js Documentation, Express.js framework and middleware.
4. TypeScript Documentation, TypeScript language reference.
5. Tailwind CSS Documentation, utility-first CSS framework.
6. Prisma Documentation, Prisma ORM and database interaction.
7. PostgreSQL Documentation, relational database concepts and SQL.
8. Passport.js Documentation, authentication middleware for Node.js.
9. Git Documentation, version control and Git workflows.
10. GitHub Documentation, repository management and collaborative development.
11. Google OAuth Documentation, OAuth 2.0 authentication concepts.